

Model Risk Lab

What a Wrong Pricing Model Costs an Options Desk — in Dollars

A market-making simulator calibrated to real SPY and NIFTY 50 data,
with a complete statistical model-validation suite

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Abstract

A market maker earns the bid–ask spread by quoting around *fair value* — but fair value comes from a model, and the model can be wrong. This project measures that danger end to end. Geometric Brownian motion (GBM) and the Merton jump-diffusion are calibrated to 17+ years of SPY and NIFTY 50 daily returns by maximum likelihood (GBM is rejected with likelihood-ratio statistics of ≈ 1627 and ≈ 1313) and to four real SPY option chains — on 16 March 2020 the options market priced $\lambda = 3.7$ crashes per year with mean size -31% . A simulated options desk then quotes a rolling three-month contract under four (true model, believed model) scenarios with price-elastic client flow. When the model matches reality the desk realises its quoted half-spread ($+0.153$ per contract against a promised 0.15 ; bootstrap 95% CI $[0.146, 0.159]$). When it prices Black–Scholes in the jump world, it *believes* it earns $+0.15$ while actually losing 10.20 per contract, and no half-spread below $\$15$ ($>170\times$ the matched desk’s break-even of 8.8 cents) makes it 95%-safe. Finally, a validation suite — Kupiec and Christoffersen VaR backtests, bootstrap edge inference, and a CUSUM monitor — detects the broken model from realised P/L alone, raising an alarm at trade 83 of 1,467 with zero false alarms on the correctly-specified desk. All data, code, figures and an interactive browser demo are in the repository. The write-up assumes bachelor’s-level probability and statistics; every model is stated with its equations, and every design choice is explained by what it does to the final number.

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1 Introduction: the business of quoting, and its hidden bet

1.1 What a market maker does

An options market maker continuously posts two prices for the same contract: a **bid** at which it will buy and an **ask** at which it will sell. If the option’s fair value is V and the desk quotes

$$\text{bid} = V - s, \quad \text{ask} = V + s, \quad (1)$$

then every client trade hands the desk $+s$ of expected profit (the *half-spread*): it sells above fair value and buys below it. Volume times s is the revenue of the business. This is real: option markets are quote-driven, and market-making firms exist on exactly this economics.

1.2 The hidden bet

The catch is the symbol V . Nobody hands the desk fair value; it must be *computed from a model* of how the underlying asset moves. The desk’s entire P/L statement therefore rests on a modelling assumption that could be wrong. The purpose of this project is to make the consequences of that assumption **visible, measurable, and statistically testable**:

1. **Calibrate** two competing models — Black–Scholes/GBM and Merton jump-diffusion — to real market data, with standard errors and formal model comparison (Section 5–6).
2. **Simulate** a market-making desk that believes one model while the world follows another (Section 7).
3. **Measure** the damage in the units risk managers use: VaR, Expected Shortfall, draw-downs, and a break-even spread — a dollar *model-risk premium* (Section 8).

4. **Validate:** treat the desk’s model as a hypothesis and reject it from realised P/L alone, using the same tests a bank’s model-risk management (MRM) team runs — Kupiec, Christoffersen, bootstrap, CUSUM (Section 9).

The one-sentence result. A desk that prices Black–Scholes in a world with COVID-sized jumps books +0.15 of “edge” per contract in its own accounting while truly earning −10.20; statistical monitoring catches the lie within weeks, and no commercially meaningful spread makes the wrong model safe.

1.3 Origin and what is new here

The seed of this project is a teaching simulator by Roman Paolucci (Quant Guild), which demonstrates spread capture vs. model risk with assumed parameters and price-blind client flow. This project rebuilds the idea as a statistics project: all parameters are *estimated* from real data with quantified uncertainty; client flow is *price-elastic* (mispricing attracts informed flow instead of being ignored); the desk quotes a *rolling constant-moneyness* contract so its economics are stationary; and the headline contribution is the *validation layer* — the machinery for detecting a wrong model from outcomes, which is the day job of model validation quants.

2 Data

- **SPY daily OHLC, 1999–2025** (6,570 days). The S&P 500 ETF — the most liquid equity underlying in the world.
- **NIFTY 50 daily OHLC, 2007–2026** (4,554 days). India’s headline index, so both a developed and an emerging market are represented.
- **SPY end-of-day option chains, 2008–2025** (24.7M quotes in the raw mirror). The repository ships (i) a month-end panel of near-90-day-to-expiry chains — 216 dates, ~145,000 quotes with bid/ask, open interest and exchange-computed implied volatilities — and (ii) four full-surface snapshots: 2017-01-03 (calm), 2020-03-16 (COVID crash, VIX 82), 2022-06-13 (bear market), 2025-06-30 (recent).
- **US risk-free rate series** for discounting choices.

Sources are public GitHub mirrors, documented with URLs and licenses in `data/README.md`; a `scripts/fetch_data.py` refreshes everything from the user’s own machine. Calibration uses log-returns $x_i = \ln(S_i/S_{i-1})$ at $\Delta t = 1/252$ years.

3 Two models of price motion

3.1 Geometric Brownian motion — the Black–Scholes world

What. The stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (2)$$

says returns have a constant expected growth rate μ and constant volatility σ , driven by Brownian motion W_t . By Itô’s lemma, log-returns over Δt are i.i.d. Gaussian:

$$\ln \frac{S_{t+\Delta t}}{S_t} \sim \mathcal{N}\left((\mu - \frac{1}{2}\sigma^2)\Delta t, \sigma^2\Delta t\right). \quad (3)$$

Why it matters. GBM is the assumption inside the Black–Scholes formula. A desk that prices with Black–Scholes is betting that returns are Gaussian: no crashes, thin tails, and continuous paths that a delta hedge can follow.

3.2 Merton jump-diffusion — a world that can gap

What. Merton (1976) superimposes crashes on the diffusion. Jumps arrive as a Poisson process N_t with intensity λ per year; each jump multiplies the price by e^Y , $Y \sim \mathcal{N}(\mu_J, \sigma_J^2)$:

$$\frac{dS_t}{S_{t-}} = (\mu - \lambda k) dt + \sigma dW_t + (e^Y - 1) dN_t, \quad k = \mathbb{E}[e^Y] - 1 = e^{\mu_J + \sigma_J^2/2} - 1. \quad (4)$$

The $-\lambda k dt$ compensator recentres the process so that μ remains the total expected growth rate — without it, adding negative jumps would silently change the drift and the comparison between models would be unfair.

Why it matters. A daily log-return under (4) is a *Poisson mixture of Gaussians*: with probability $e^{-\lambda \Delta t} (\lambda \Delta t)^n / n!$ the day contains n jumps, and conditionally the return is Gaussian with mean shifted by $n\mu_J$ and variance inflated by $n\sigma_J^2$. Mixing Gaussians of different variances is precisely what generates **fat tails and skew** — the two features of real returns that Gaussians cannot produce.

4 Pricing under each model

4.1 Black–Scholes

Under GBM with risk-free rate r the European call price is

$$C_{BS} = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2), \quad d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}, \quad (5)$$

with Φ the standard normal CDF; puts follow by put–call parity ($C - P = S_0 - Ke^{-rT}$, verified in unit tests to 10^{-10}). The hedge ratio is $\Delta_{\text{call}} = \Phi(d_1)$, $\Delta_{\text{put}} = \Phi(d_1) - 1$.

4.2 Merton: a Poisson-weighted average of Black–Scholes prices

Conditional on n jumps during the option’s life the terminal log-price is still Gaussian, so the price is a mixture of Black–Scholes prices with jump-adjusted inputs:

$$C_M = \sum_{n=0}^{\infty} \frac{e^{-\lambda'T} (\lambda'T)^n}{n!} C_{BS}(S_0, K, T; r_n, \sigma_n), \quad \lambda' = \lambda(1 + k), \quad (6)$$

$$\sigma_n^2 = \sigma^2 + \frac{n\sigma_J^2}{T}, \quad r_n = r - \lambda k + \frac{n(\mu_J + \sigma_J^2/2)}{T}.$$

The series is truncated at 60 terms; the neglected Poisson tail is below 10^{-40} for all parameters used here. The Merton delta is the same weighted average of Black–Scholes deltas. **Impact:** because option payoffs are convex, the fat tails raise the price — for identical diffusive σ , $C_M > C_{BS}$. That gap, evaluated at calibrated parameters, is the *mispricing per contract* that a Black–Scholes desk gives away in a jump world.

5 Calibration I: the physical measure

5.1 GBM by closed-form maximum likelihood

With Gaussian log-returns, maximising the likelihood gives the sample mean $\hat{m}$ and (MLE) variance $\hat{v}$ of daily returns, mapped to annual units:

$$\hat{\sigma} = \sqrt{\hat{v}/\Delta t}, \quad \hat{\mu} = \frac{\hat{m}}{\Delta t} + \frac{\hat{\sigma}^2}{2}. \quad (7)$$

Standard errors use the exact Gaussian Fisher information, $\text{Var}(\hat{m}) = v/n$ and $\text{Var}(\hat{v}) = 2v^2/n$, propagated by the delta method. **An honest disclosure:** with $n \approx 4,500$ days, $\hat{\sigma}$ is pinned to three decimals but $\hat{\mu}$ carries a standard error of $\sim 4.5\%$ *per year* — drift is nearly unlearnable from daily data. This is why the desk's P/L design leans on delta-hedged positions, whose expectation is drift-free (Section 7).

5.2 Merton by numerical maximum likelihood

The daily-return density is the truncated Poisson mixture

$$f(x) = \sum_{n=0}^8 \frac{e^{-\lambda\Delta t} (\lambda\Delta t)^n}{n!} \varphi\left(x; \left(\mu - \frac{\sigma^2}{2} - \lambda k\right)\Delta t + n\mu_J, \sigma^2\Delta t + n\sigma_J^2\right), \quad (8)$$

maximised over $(\mu, \sigma, \lambda, \mu_J, \sigma_J)$ with positivity enforced by log-parameterisation. Because jumps and diffusion can partially trade off, the likelihood is multi-modal: we use multi-start Nelder–Mead followed by BFGS polishing. Standard errors come from the *observed information* — the inverse of the numerically-differentiated Hessian of $-\ln L$ at the optimum — mapped through the log-links by the delta method. The estimator is verified on synthetic data (parameters recovered within sampling error) before touching real returns.

5.3 Model comparison: LR, AIC, BIC

The models are nested (Merton $\rightarrow$ GBM as $\lambda \rightarrow 0$), so we report

$$LR = 2(\ell_M - \ell_G), \quad AIC = 2k - 2\ell, \quad BIC = k \ln n - 2\ell. \quad (9)$$

Caveat stated openly: the null $\lambda = 0$ lies on the boundary of the parameter space, so the usual χ_3^2 reference is conservative (the true null distribution is a mixture with more mass near zero). With the observed statistics this subtlety cannot change the verdict.

Table 1: Physical-measure MLE on daily log-returns ($\pm$ asymptotic SE). χ_3^2 95% critical value: 7.81.

	SPY (2008–2025)	NIFTY 50 (2007–2026)
n (days)	4,504	4,549
GBM $\hat{\sigma}$	0.2001 ± 0.0021	0.2071 ± 0.0022
Merton $\hat{\sigma}$ (diffusive)	0.0979 ± 0.0039	0.1223 ± 0.0038
Merton $\hat{\lambda}$ (jumps/yr)	87.9 ± 10.6	59.8 ± 8.4
Merton $\hat{\mu}_J$	-0.0026 ± 0.0006	-0.0015 ± 0.0008
Merton $\hat{\sigma}_J$	0.0175 ± 0.0010	0.0207 ± 0.0013
LR vs GBM	1627	1313
ΔBIC (GBM – Merton)	1602	1288

Reading Table 1. On both markets the data split the GBM volatility into a much smaller diffusive part plus a stream of frequent small mostly-negative jumps (≈ 88 /year for SPY of

mean -0.26%). The likelihood improvement is overwhelming: $\Delta BIC > 10$ is conventionally “decisive”; here it is above 1,200. Figures 1 and 2 show *where* the Gaussian fails: the tails. SPY’s worst in-sample day (-11.6%) is a $\approx 9\sigma$ event under fitted GBM — effectively impossible — while the Merton mixture assigns it sensible probability.

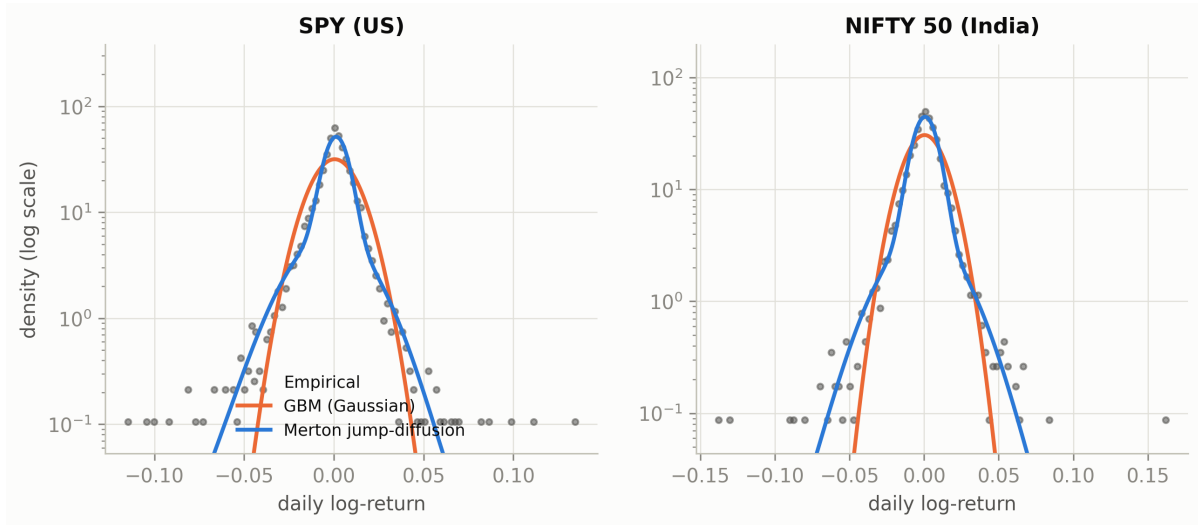


Figure 1: Daily-return densities on a log scale: empirical (grey points), fitted GBM (orange), fitted Merton (blue). The Gaussian collapses many decades too fast; the jump mixture tracks the empirical tails on both markets.

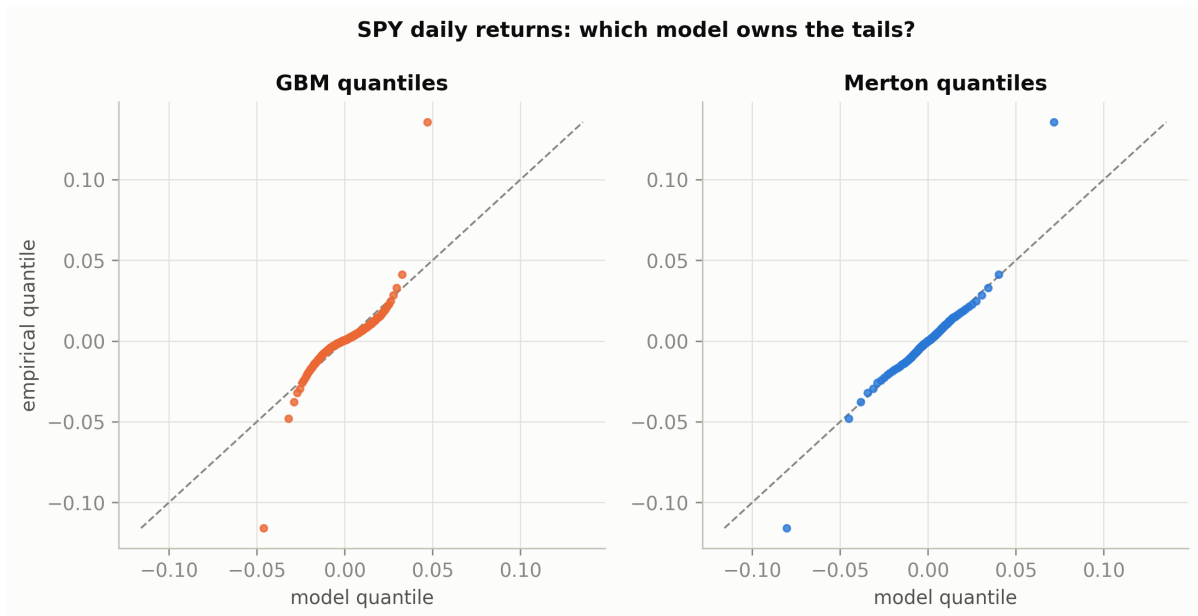


Figure 2: QQ plots of SPY daily returns against each fitted model. Against GBM the empirical quantiles peel violently off the diagonal in both tails; against Merton they hug it.

6 Calibration II: the risk-neutral measure

Option prices reveal the *market’s* model. Quoting each option as its Black–Scholes **implied volatility** (the σ that reproduces its price), a Gaussian world would produce a flat line across

strikes; real markets produce a **skew** — out-of-the-money puts trade at much higher implied vols, because the market pays up for crash insurance.

For each snapshot we take the expiry nearest 90 days, keep OTM quotes with positive bids (the liquid side of the smile), compute mid-price implied vols by Brent root-finding (quotes violating no-arbitrage bounds are dropped), and fit Merton parameters by weighted least squares on the smile:

$$\min_{\sigma, \lambda, \mu_J, \sigma_J} \sum_i w_i (IV_M(K_i) - IV_{\text{mkt}}(K_i))^2, \quad w_i \propto \exp \left[-\frac{1}{2} \left(\frac{\ln(K_i/S)}{0.15} \right)^2 \right], \quad (10)$$

ATM-centred weights because at-the-money quotes carry the most information per unit of noise (highest vega). For speed the optimiser uses the standard first-order equivalence $\Delta IV \approx \Delta \text{price}/\text{vega}$ with vega frozen at market IVs; the reported fit quality is recomputed by exact inversion.

Table 2: Risk-neutral Merton calibrated to four real SPY chains (≈ 90 DTE, OTM quotes).

Snapshot	Regime	ATM IV	σ	λ/yr	μ_J	σ_J	RMSE (vol pts)
2017-01-03	calm bull	13.5%	0.068	0.92	−0.106	0.100	1.27
2020-03-16	COVID crash	64.2%	0.116	3.68	−0.308	0.190	0.74
2022-06-13	bear market	30.1%	0.128	2.00	−0.162	0.136	0.67
2025-06-30	recent	14.3%	0.080	0.81	−0.138	0.115	1.05

Reading Table 2. Even the calmest market prices roughly one -10% jump per year — the smile *never* believes Black–Scholes. In March 2020 the market priced $\lambda = 3.7$ crashes/year of mean -31% . A 4-parameter model reproduces 40+ vol-point skews to within about one vol point (Figure 3). These COVID-chain parameters — a jump world someone actually paid for — define the “true world” of the mismatch experiments.

7 The desk: generating market-making P/L under model risk

7.1 Mechanics

Time advances in ticks ($dt = 1/250$ years). At each tick the desk quotes a *fresh* option with constant maturity $T = 3$ months and strike at current spot ($K_t = S_t$): rolling constant moneyness keeps the business stationary — a fixed strike would drift out of the money as the underlying trends, silently shrinking both risk and revenue and contaminating every downstream statistic (this exact artefact was caught by a failing coverage backtest during development and is documented in the repo history). Around its believed fair value it posts bid $= V_{\text{mod}} - s$, ask $= V_{\text{mod}} + s$.

When a client trades, the contract’s P/L is **realised at the fill**: the underlying is simulated over the option’s life under the *true* model (GBM or Merton), while the desk delta-hedges at 30 rebalance dates using *its own* model’s delta, financing the hedge at r ; premium, hedge cash flows and terminal payoff are discounted to the fill date. Realising at the fill keeps the equity curve a story about *pricing edge*, not about an ever-growing open inventory position.

7.2 Why a hedged option earns “premium minus fair value”

For a delta-hedged short option rebalanced continuously under the model’s own dynamics, the dW terms cancel and the hedged portfolio grows at r ; the discounted P/L of (sell at premium p , hedge to expiry) converges to

$$\text{P/L} \longrightarrow p - V_{\text{fair}}, \quad (11)$$

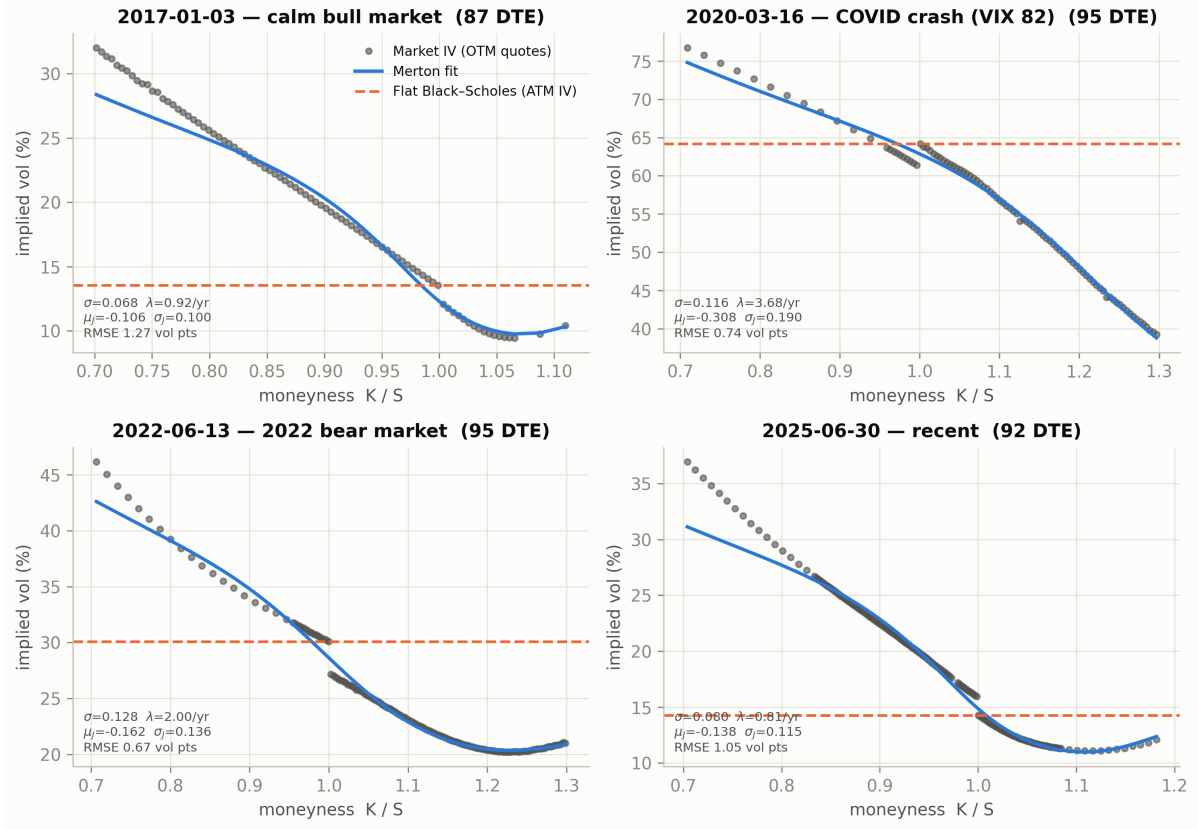


Figure 3: Real SPY implied-vol skews (grey), Merton fits (blue), and the flat smile Black-Scholes implies (orange dashed), on four dates spanning calm to crisis.

independent of the drift μ . Selling at $V + s$ or buying at $V - s$ then banks $+s$ per contract in expectation. Model error breaks both legs of this argument simultaneously: the quotes centre on the wrong V (a systematic, sign-definite error), and the hedge follows the wrong delta — worse, *no* delta hedge can follow a jump, because the price gaps a finite amount in an instant. Two further honest caveats: with 30 discrete rebalances the cancellation in (11) is approximate (mean-zero noise remains), and under jumps the market is *incomplete* — even the right-model desk retains undiversifiable jump risk plus a small residual drift exposure, which is why its realised edge in Table 3 (0.194) sits slightly above the quoted 0.15 rather than exactly on it.

7.3 Elastic client flow: mispricing attracts sharks

Real clients compare quotes to value. Each side’s fill probability is scaled by how bad the quote is for the client, in units of half-spread against *true* fair value:

$$p_{\text{fill}} = \min\left(0.98, p_0 e^{-\eta \cdot \text{excess}}\right), \quad \text{excess}_{\text{ask}} = \frac{\text{ask} - V_{\text{true}}}{s} - 1, \quad (12)$$

capped at $e^2 \times$ the base rate (client arrival is finite even when the desk gives money away). A fairly-priced quote trades at the base rate; a rich quote loses business; a cheap quote is run over by informed flow — adverse selection. Flow is put-heavy (base $\mathbb{P}(\text{client buys}) = 0.5$ vs $\mathbb{P}(\text{client sells}) = 0.1$ per tick), matching the real net demand for index crash protection, so a desk that underprices puts ends up systematically *short* crash insurance. Without elasticity ($\eta = 0$) the simulator reproduces the textbook version where an overpricing desk looks brilliant; with it, both directions of mispricing are punished — which is what Table 3 shows.

7.4 The four scenarios

All use $S_0 = 100$, ATM puts, $T = 0.25$, $r = 2\%$, $\mu = 6\%$, $s = 0.15$, 250 ticks (one year), seeds $0..n-1$:

- **Matched**: world GBM ($\sigma = 0.20$, the SPY fit); desk prices Black–Scholes with the same σ .
- **Steamroller**: world Merton with the COVID-chain parameters ($\sigma = 0.116$, $\lambda = 3.68$, $\mu_J = -0.308$, $\sigma_J = 0.19$); desk prices Black–Scholes with only the diffusive σ .
- **Right model**: world as above; desk prices Merton with the true parameters.
- **Overcautious**: world GBM; desk prices Merton with COVID jumps that never come.

8 Results I: P/L distributions and the price of model risk

Table 3: Desk experiments across seeds. Edge CI is a 95% percentile bootstrap on per-seed average realised edge per contract. Negative VaR/ES means the 5% tail is still a profit.

Scenario	Seeds	Believed edge	Realised edge [95% CI]	Mean P/L	VaR _{5%}	ES _{5%}
Matched	300	+0.15	+0.153 [0.146, 0.159]	+22.8	−9.4	−4.0
Right model	200	+0.15	+0.194 [0.064, 0.324]	+28.3	207.9	289.7
Steamroller	300	+0.15	−10.20 [−10.62, −9.78]	−2497.8	4012.5	4466.9
Overcautious	200	+0.15	−9.08 [−9.23, −8.92]	−1671.1	2044.7	2127.9

Reading Table 3. Three lessons. (i) The matched desk delivers exactly what it promises — realised edge statistically indistinguishable from the quoted half-spread. (ii) The mismatched desks lose catastrophically *in both directions*: underpricing is picked off by buyers of cheap crash insurance (steamroller), overpricing is picked off by sellers dumping overpriced options on the desk’s rich bid (overcautious). Model risk is not “being too cheap”; it is *being wrong in either direction when the other side knows better*. (iii) Even the right-model desk in a jump world carries much fatter P/L tails than the matched desk (incomplete market: jumps cannot be hedged away), which is a fact about markets, not a bug.

8.1 The model-risk premium

Re-running the desk over a grid of half-spreads traces the loss-probability curve $s \mapsto \mathbb{P}(\text{P/L}(s) < 0)$ (Figure 6). Define the break-even spread as the smallest s with $\mathbb{P}(\text{loss}) \leq 5\%$:

Matched desk: breaks even at $s^* = 8.8$ cents.

Mispriced desk: no break-even below \$15 — more than $170\times$ wider. With informed (elastic) flow, widening the quote throttles the business faster than it buys safety: the desk cannot spread its way out of a wrong model.

This is the project’s answer to “what does model risk cost”: for this desk, the model-risk premium is not a few extra cents of spread — it is the difference between a viable business and no business at all.

9 Results II: validation — catching the wrong model from P/L alone

A model validator does not get to observe “the true model”; they observe *outcomes*. This section runs the standard MRM toolkit on the desk’s realised P/L, on a six-year run (1,500

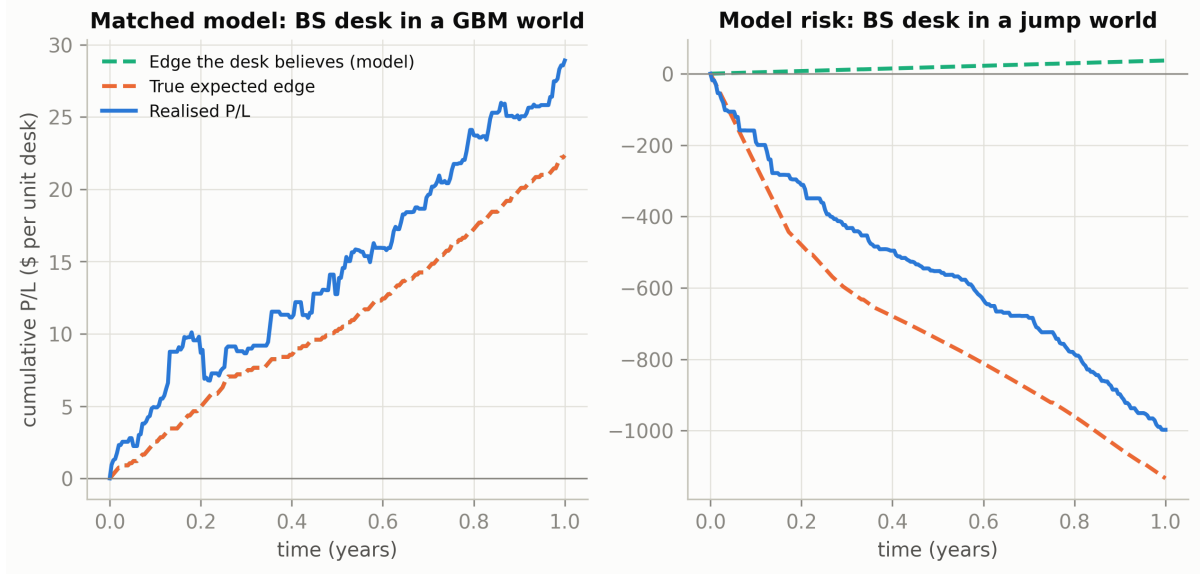


Figure 4: One seed, two worlds. Left: matched desk — realised P/L rides the believed and true edge lines. Right: steamroller — the desk’s own accounting (teal, top) reports steady profit while true expectation and realised P/L collapse. The vertical gap *is* model risk in dollars.

ticks) for test power. To keep the P/L series stationary, all per-tick quantities are scaled by the spot level (risk per unit of notional — an ATM option’s dollar value is proportional to S).

9.1 Kupiec’s proportion-of-failures (POF) test

The desk forecasts its own 5% one-tick VaR by simulating its *believed* world (the standard model-based VaR practice). If the model is right, losses should exceed VaR on $\alpha = 5\%$ of ticks, independently. Observing x exceptions in n ticks:

$$LR_{\text{pof}} = -2 \ln \frac{(1 - \alpha)^{n-x} \alpha^x}{(1 - \hat{\pi})^{n-x} \hat{\pi}^x} \stackrel{H_0}{\sim} \chi_1^2, \quad \hat{\pi} = x/n. \quad (13)$$

9.2 Christoffersen’s independence and conditional coverage

Correct frequency is not enough — exceptions must not *cluster*. Model the exception indicator as a two-state Markov chain with transition counts n_{ij} and conditional exception probabilities π_0, π_1 ; test $\pi_0 = \pi_1$:

$$LR_{\text{ind}} = -2 \ln \frac{(1 - \hat{\pi})^{n_{00}+n_{10}} \hat{\pi}^{n_{01}+n_{11}}}{(1 - \hat{\pi}_0)^{n_{00}} \hat{\pi}_0^{n_{01}} (1 - \hat{\pi}_1)^{n_{10}} \hat{\pi}_1^{n_{11}}} \stackrel{H_0}{\sim} \chi_1^2, \quad LR_{\text{cc}} = LR_{\text{pof}} + LR_{\text{ind}} \stackrel{H_0}{\sim} \chi_2^2. \quad (14)$$

Clustered exceptions are the signature of dynamics the model does not have — here, jumps.

9.3 Bootstrap inference on realised edge

Per-seed average edges are i.i.d. by construction, so a percentile bootstrap (10,000 resamples) gives honest confidence intervals for mean edge, reported in Table 3, alongside a one-sample t -test of H_0 : mean edge = 0.

9.4 CUSUM: the early-warning monitor

Waiting a year for a backtest is a luxury; a desk bleeding edge should be caught in weeks. The one-sided CUSUM tracks cumulative under-delivery of the promised edge m on spot-scaled

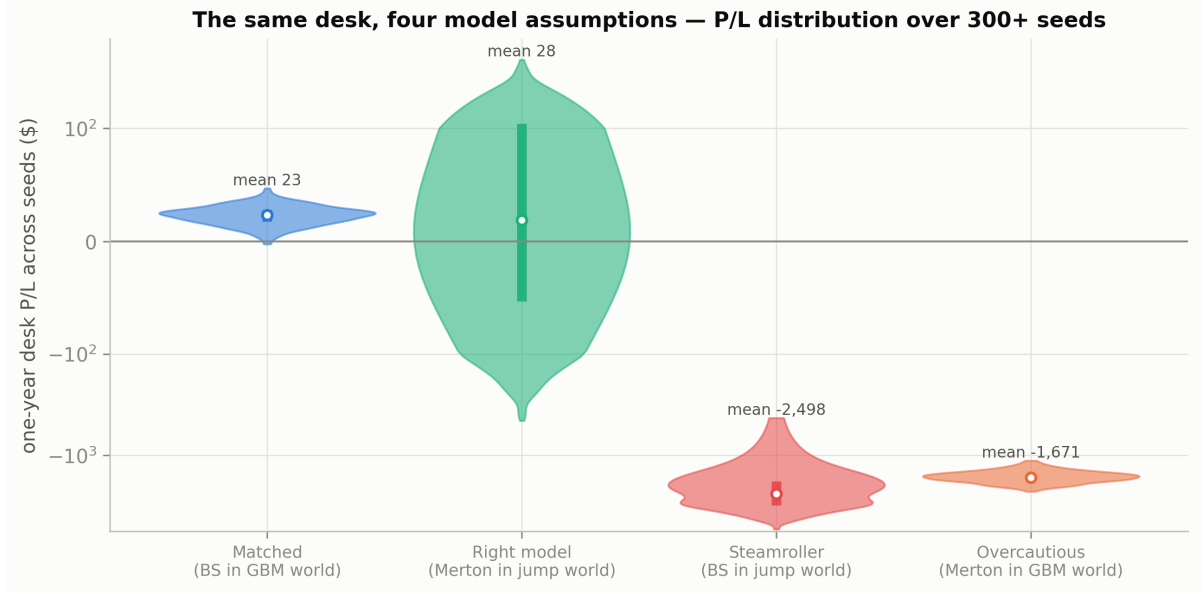


Figure 5: One-year P/L across seeds, all four scenarios (symlog scale).

per-trade residuals X_t :

$$C_0 = 0, \quad C_t = \max(0, C_{t-1} + (m - X_t) - k), \quad \text{alarm when } C_t > h, \quad (15)$$

with allowance $k = 0.5\hat{\sigma}$ and threshold $h = 8\hat{\sigma}$, control limits calibrated on a burn-in window that is excluded from monitoring (you do not raise alarms on the data used to set the limits).

Table 4: Validation results on six-year runs (1,500 ticks; expected exceptions 75).

Test	Matched desk	Mispriced desk	Verdict
Kupiec POF	83 exc., $p = 0.351$	811 exc., $p < 10^{-16}$	right model passes; wrong fails wrong model's exceptions cluster detection in ~ 3 weeks
Christoffersen CC	$p = 0.619$	$p < 10^{-16}$	
CUSUM	no alarm in 903 trades	alarm at trade 83 of 1,467	

The validator's takeaway. The wrong model is not merely unprofitable — it is *statistically detectable* from realised P/L, quickly, with tests whose false-alarm behaviour is verified on the correctly-specified desk. This is the entire logic of model risk management, demonstrated end to end on one desk.

10 Limitations, stated plainly

- **Stylised microstructure.** One contract, Bernoulli client arrivals, a reduced-form elasticity (12) rather than an order book or a utility-based client model.
- **Static parameters within runs.** No stochastic volatility, no parameter drift; the “truth” inside each experiment is a fixed Merton world. Real model risk includes the model being wrong *in form*, not only in parameters — our mismatch captures exactly one axis of that.
- **Boundary null in the LR test** (Section 5); conclusions are robust here but the caveat matters at smaller samples.

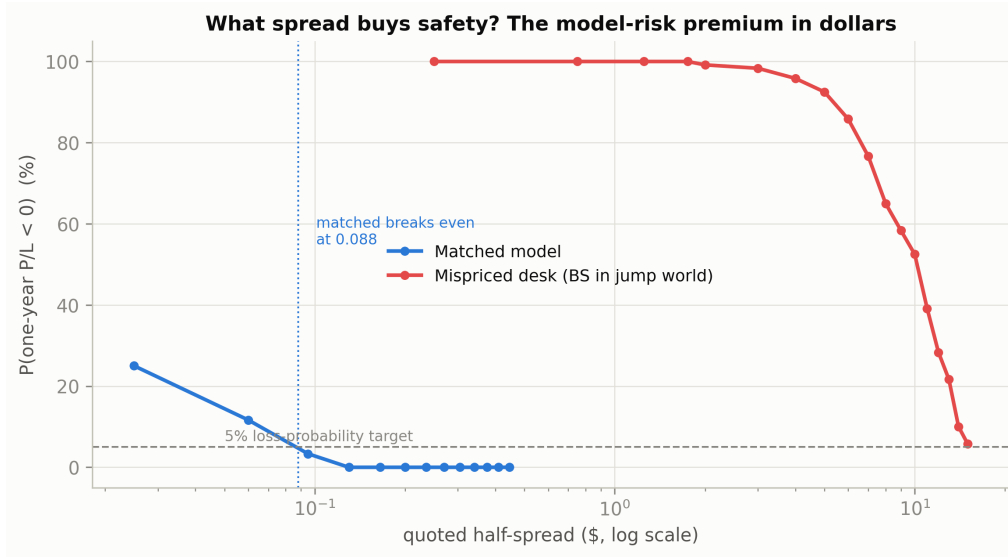


Figure 6: Loss probability vs quoted half-spread (log scale).

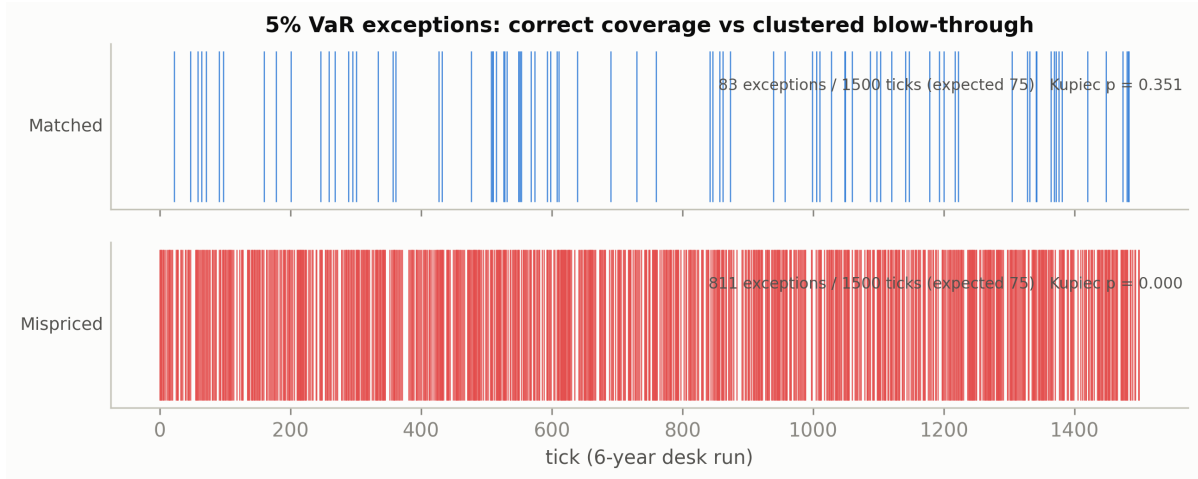


Figure 7: 5% VaR exception timelines. Matched: 83 scattered exceptions vs 75 expected. Mispriced: 811, clustered into jump episodes.

- **Incomplete-market subtleties** under jumps: hedged P/L is not fully drift-free, and jump risk premia (difference between P - and Q -measure jump parameters) are real economics that the right-model desk partially collects; a full treatment would separate them explicitly.
- **Data vintage.** Chains end December 2025; NIFTY options are not in the free mirror used (the repo ships a fetch script for NSE data to extend the Indian leg).

None of these change the qualitative conclusions; each is a natural extension.

11 Reproducibility

Everything regenerates from two commands:

```
python scripts/run_experiments.py → python scripts/make_figures.py
```

Deterministic seeds throughout; 15 unit tests (put-call parity, delta vs finite differences, Merton → BS convergence, MLE recovery on synthetic data, matched desk realising its half-

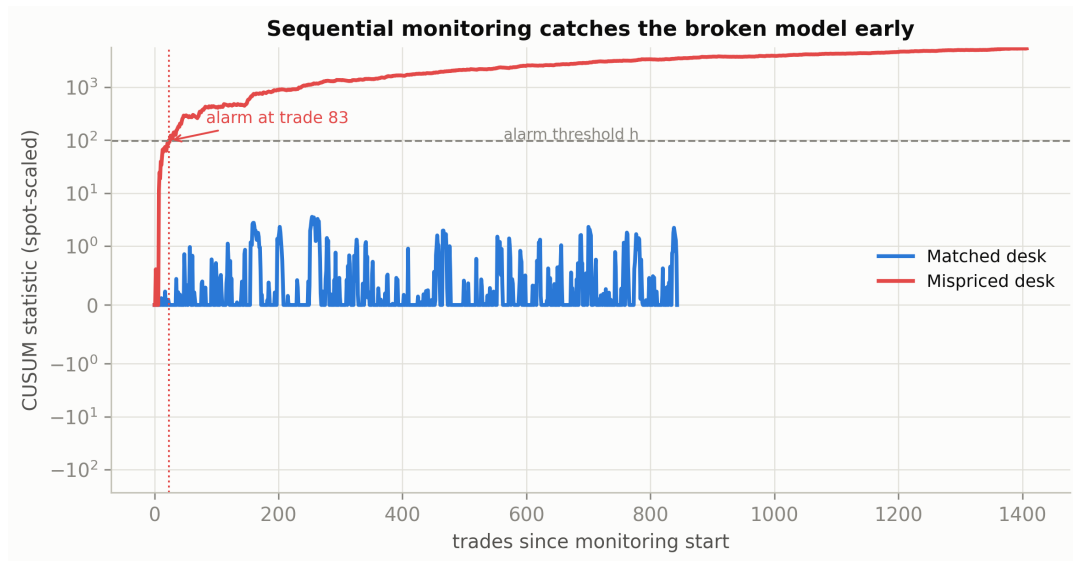


Figure 8: CUSUM paths. The matched desk’s statistic breathes below threshold for 900+ trades (zero false alarms); the mispriced desk crosses at trade 83.

spread, Kupiec/Christoffersen/CUSUM behaviour on synthetic series) run via `python -m pytest tests/`. The interactive website ports the same engine to JavaScript with a seeded RNG. Repository: <https://github.com/Arijitray2/model-risk-lab>.

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